The Vita stack — 2026 industry consensus + decision

> *Authored 2026-05-03 in response to the Substrate v0.1 scope question: what language, what hosting, what database for the platform. Decision is **YC-consensus stack** (~60-70% of new agent startups in 2026).*

What new agent companies actually shipped this year

Choice	Industry default 2026	Source signal
Primary language	**TypeScript**	60-70% of YC X25 agent batch · #1 on GitHub (2.63M contributors)
Frontend framework	**React + Tailwind**	44.7% adoption · default for Vercel / v0 / Lovable / Bolt
API layer	**Hono**	Lightweight, runs on Node + Workers + Vercel + Bun without rewrites
Database	**PostgreSQL**	55.6% adoption · usually managed via Supabase or Neon
Frontend hosting	**Vercel**	Uncontested for Next.js
Backend hosting	**Railway**	Long-running, persistent containers, no timeout ceiling
Total infra cost (pre-revenue)	**< $100/mo**	Vercel Pro $20 + Railway $20-40 + Supabase $25
Rust's role	**Build tooling only**	Turbopack, Rspack, Biome, oxlint — under the hood, not in app code
Model layer	**Multi-model, swappable**	Cursor / Claude Code / Augment all let you flip Claude / GPT / Gemini / Grok mid-task

The Vita stack

```
TypeScript monorepo · pnpm + Turborepo
│
├── apps/
│   └── web/               → Vercel             Next.js 16 + React 19 + Tailwind v4
│
├── packages/
│   ├── spec/              → npm publishable    JSON Schema → typed manifest contract
│   ├── runtime/           → Railway            dispatch · audit · policy · sign
│   ├── orchestrator/      → Railway            agent loop · planning · streaming
│   ├── ontology/          → consumed in-place  MD axioms compiled to TS exports
│   └── ui/                → consumed in-place  shadcn-style primitives for spaces
│
├── manifests/             → versioned in repo  plunet/0.1.0.yaml, xero/0.1.0.yaml, …
│
├── adapters/              → Railway            http · browser (Playwright) · desktop · hil · mcp
│
└── infra/
    └── supabase/          → Supabase us-west-1  one project · RLS multi-tenant
```

Hosting topology

Service	Where	Why
`viter.ai` (Next.js UI)	**Vercel**	Their product. Edge cache. Per-branch previews.
`api.viter.ai` (chat streaming, signed URLs)	**Vercel Functions (Fluid)**	800s ceiling enough for SSE streams. `waitUntil()` for post-response work.
`substrate.viter.ai` (runtime + adapters)	**Railway**	Always-on. No timeout ceiling. Browser sessions stay warm for Playwright.
Data layer (manifests · audit · chat · wiki · finance)	**Supabase** us-west-1	One Postgres. RLS for tenancy. Already running.

What this is NOT

- Not a Python codebase. Yitzchak's `Knowledge-Agent` (Python/FastAPI) is the largest single migration cost — ~2 weeks to port the chat / wiki / agent loop into `packages/orchestrator/` as TypeScript. Real cost. Worth it for monorepo type-safety + 2-person team velocity.
- Not Rust at the app layer. Rust is real for *agent infrastructure substrate* (16× growth in 2026 OSS) but the team-velocity cost (2-4 week ramp) eats half the v0.1 window. Document Rust as the **v0.2 escape valve** for hot adapters once a real bottleneck surfaces.
- Not split across two Supabase projects. Knowledge-Agent's Tokyo project (`vwqalkghhdgjumjdgtpd`) gets retired; data migrates to viter's us-west project (`mcghcqbjtwtkdyezcswr`). Wrong region for an EU/IL team anyway.
- Not Cloudflare Workers in v0.1. Add only if a specific adapter genuinely needs global edge — overkill for one design partner (Insperanto) starting May 1.

The Vita decision in one paragraph

Build Vita on the YC-consensus stack — TypeScript monorepo, Vercel for UI, Railway for substrate runtime + Playwright adapters, Supabase for one-Postgres-everything. This is what 60-70% of new agent startups picked in 2026, costs under $100/month pre-revenue, and is uncontested at this scale. Don't fight infra choices. Fight where it actually matters: the **ontology** (Karp moat — vertical worldview), the **manifest format** (substrate moat — typed contract that competitors can't replicate cheaply), and the **Persofi worldview** (the patent-as-job axiom Shaul's father surfaced — the actual product). Infra is commodity in 2026. Your moat is two layers up.

Sources

- 60–70% of YC X25 Agent Startups Are Using TypeScript
- The Best Startup Tech Stack in 2026 — Our Opinionated Guide
- My 2026 Tech Stack Predictions — YUV.AI
- YC W26 Batch Breakdown: 199 Companies
- Railway vs. Vercel — Railway Docs
- AI Agent Hosting in 2026: Render vs Railway vs Serverless Stacks
- Vercel Fluid Compute: Serverless Functions Guide (2026)
- The Rust Shift: How GitHub's AI Agent Infrastructure Changed Languages
- TypeScript Rising: Replacing Python in Multi-Agent AI Systems
- 12 Best Agentic Engineering Platforms and AI Tools (2026)

Related

- `00-dependency-map.md` — the layering above the stack
- `00-dependency-map-debt.md` — broken contracts in the existing code
- `06-substrate-vs-ontology.md` — why ontology sits above substrate
- `07-spaces.md` — the role-shaped UI that consumes substrate via ontology